

Nicholas Zhang

(604)-997-0377 | nicholas.zhang4008@gmail.com | Coquitlam, BC | github.com/NicholasZ4008 | nicholaszhang.ca

TECHNICAL SKILLS

Languages: C#, Java, JavaScript/TypeScript, Python, SQL, HTML/CSS

Frameworks & Tools: Unity, .NET/ASP.NET Core, React, Next.js, Node.js, Entity Framework Core, Jest, Docker, Git, Azure DevOps, PostgreSQL, SQL Server, Jira, Coolify

EXPERIENCE

Web Application Developer Co-op

May 2025 – Jan 2026

Public Services and Procurement Canada, Government of Canada

Vancouver, BC

- Built and maintained ASP.NET Core MVC applications with Entity Framework Core and SQL Server for federal maritime infrastructure, including a vessel booking system for the largest commercial drydock on the West Coast
- Implemented role-based access control and comprehensive form validation for booking workflows, coordinating requirements directly with client subject matter experts
- Refactored controllers and enhanced RESTful API endpoints for a hydrographic survey visualization system used to calculate safe navigation drafts on the Fraser River
- Built client-side interfaces for a maritime navigation reporting system (Avadepth) using vanilla JavaScript, implementing JSON-driven dynamic menus and date validation for tide and depth calculations

Project Manager and Web Developer Co-op

Jan 2025 – May 2025

Provincial Health Services Authority

Langley, BC

- Developed web applications using ASP.NET and React with TypeScript for an ongoing ERP integration project
- Wrote unit tests with Jest, improving code coverage and reducing post-release defects

Frontend Test Engineer Co-op

Jan 2023 – May 2023

Epic Safety Inc.

Vancouver, BC

- Developed Jest test suites for React components, increasing code coverage by 40%

Vice President of Technology

September 2023 – Present

The Forum SFU | theforumuniversity.com

Burnaby, BC

- Lead a development team building and maintaining a responsive club website (Next.js, Tailwind CSS) serving 1,000+ guests at flagship events
- Implemented secure email signup with ASP.NET backend and PostgreSQL, self-hosted on a personal Ubuntu server via Coolify with Docker containers

PROJECTS

Mon Battlers | *Unity, C#, ComfyUI*

March 2026 – Present

- Developing a Pokémon-inspired turn-based monster battler in Unity as a game development portfolio piece
- Architected a data-driven monster system using ScriptableObjects for species data and a plain C# `BattleMonster` class for runtime state, cleanly separating static and mutable concerns
- Implemented an FSM-based battle system (`PlayerTurn` → `EnemyTurn` → `ApplyEffects` → `CheckFaint`) with a type effectiveness matrix and damage formula

Digital Diversity Map | *JavaScript, PHP, SQL Server, Google Maps API*

Jan 2025 – May 2025

- Independently led full-stack development of an interactive web application for 500+ team members to share cultural connections, now used by the organization's Equity, Diversity & Inclusion team
- Integrated Google Maps API with Places Autocomplete, marker clustering, and real-time data persistence using PHP/SQL Server within a Drupal CMS
- Implemented WCAG 2.1 accessibility compliance and responsive design following Government of Canada standards

Personal Server | *Ubuntu Server LTS, Coolify, Docker*

August 2024

- Configured and secured a Lenovo TS440 with Ubuntu Server LTS to self-host production web applications, managing deployments through Coolify with Docker containers

EDUCATION

Simon Fraser University

B.Sc. Computer Science

Vancouver, British Columbia

September 2026